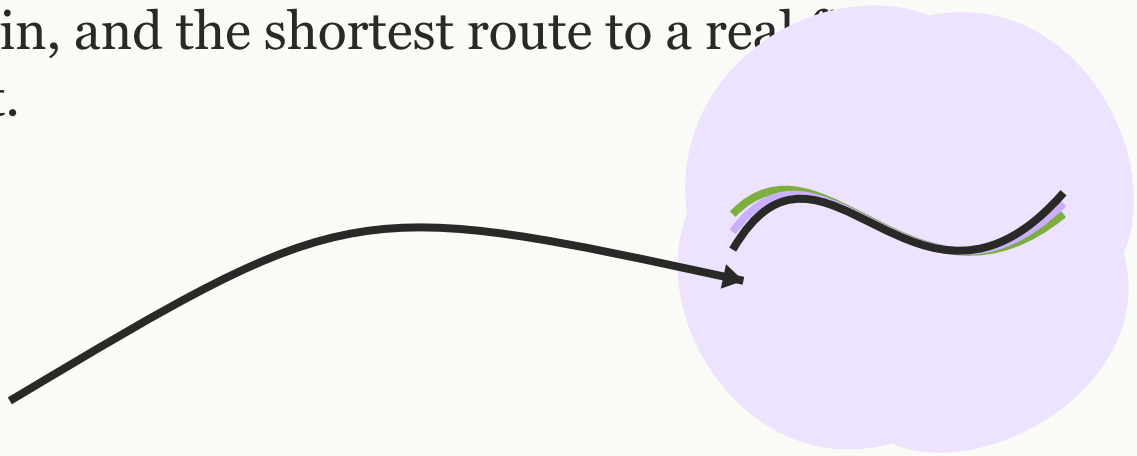
PRINTABLE FIELD GUIDE

THE DJ ENGINE, EXPLAINED.

What exists, why it works, what is still uncertain, and the shortest route to a rea[illegible] product.



This document is the compact mental model of the project. It explains the product goal, the analysis and rendering pipelines, the first successful listening result, the quality of the implementation, the main risks, and the next build sequence.

It is designed to be read from beginning to end in one sitting. File paths appear only to show where the implementation lives; the core knowledge is explained here rather than delegated to other documents.

Current through decision D038 and experiment E026 - 06 September 2026.

THE SHORT VERSION

A working manual DJ engine exists. The automatic product does not - yet.

The overall goal

Given a folder of owned music and a natural-language request, the product should choose suitable tracks, arrange a coherent journey, plan transitions, and render a finished mix. A language model may propose the plan, but deterministic code must remain responsible for validation, safety, and audio execution.

What works today

The protected ten-track house collection has been analyzed. The engine can score compatibility, expose compact planning windows, validate a typed manual MixPlan, and render a complete WAV with an audit manifest. Source files remain unchanged and are verified by hash.

The first real success

The second three-track render uses two 16-bar house_eq_v1 transitions. Each overlap lasts 29.09 seconds, and the measured adjacent-grid drift is 10 ms and 0 ms. The user accepted the duration and downbeat alignment as the first working baseline.

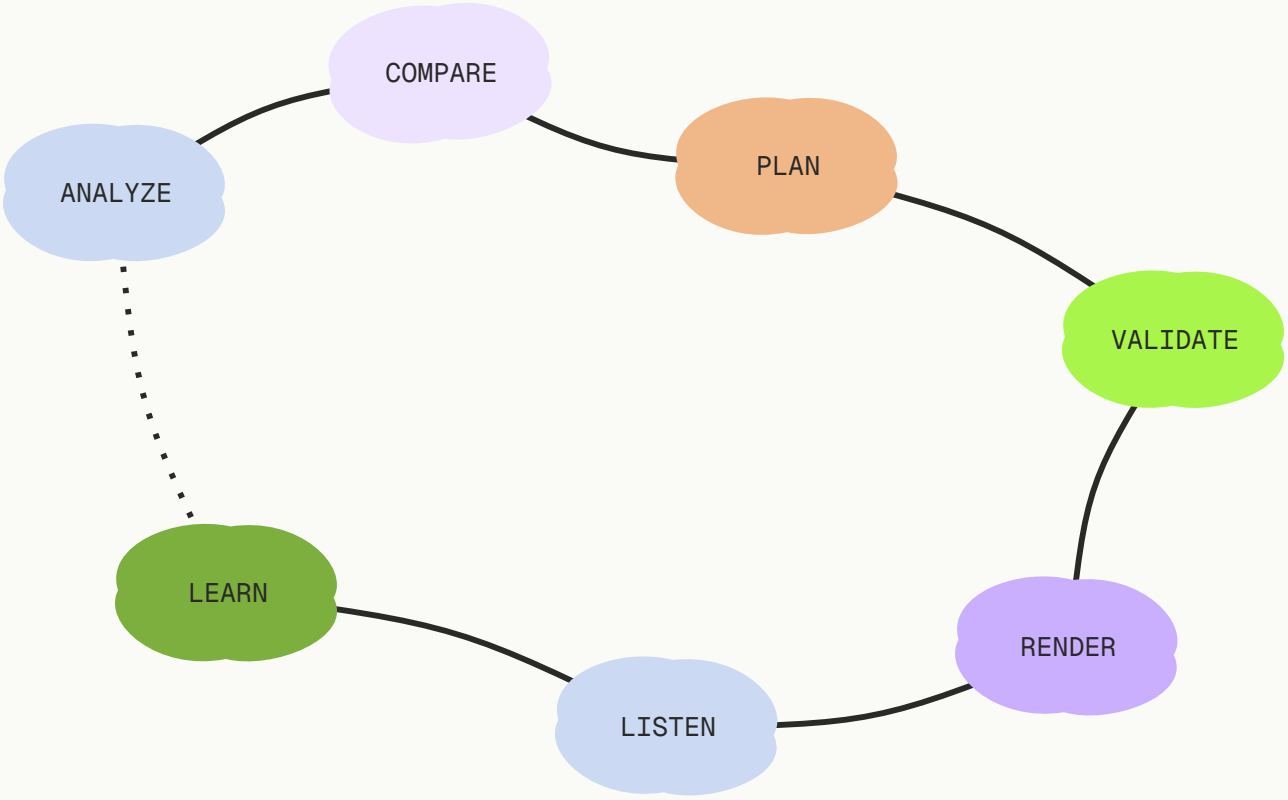
What comes next

The accepted render peaks at +1.04 dBTP, so it is not yet delivery-safe. The fastest responsible path is to add deterministic final-bus headroom or limiting, then build a five- or six-track manual set and evaluate it with a repeatable listening rubric. Automatic planning comes after several reviewed manual plans work.

ONE-LINE VERDICT

Technically strong for an early MVP, musically promising, and operationally disciplined - but still narrow in listening evidence and unfinished as an automatic product.

THE PRODUCT IS A LOOP, NOT A SINGLE MODEL.



The central design decision is separation of authority. Analysis produces structured evidence. Scoring and planning decide what a set could do. Validation rejects unsafe or inconsistent plans. Rendering performs only bounded operations. Listening then supplies the human evidence needed to revise the rules.

> This separation is why the future language-model layer is comparatively safe: it will output the same declarative MixPlan that a person writes today. It will not receive permission to execute arbitrary audio code, change source files, bypass limits, or invent facts.

STEP 1 COMPLETE STEP 2 CORE WORKS STEP 3 NOT STARTED

WHAT EXISTS RIGHT NOW



The current artifact contains ten of ten authorized house tracks with zero collection failures: My Paradise, For The Record, Fantazia, Love State, Certified, Rock The House, Dj Yellow, Wildest Desire, Lemon8, and Life Is Trippy. Together they produce 232 detected sections and 7,384 loop candidates in a 4.1 MB JSON collection.

Implemented

Step 1 analysis is complete for this corpus. Step 2A scoring is implemented. Step 2B includes planner lookups, bounded clip operations, a strict manual MixPlan validator, complete set rendering, manifest generation, and downbeat-click diagnostics.

Heard evidence

The accepted v2 plan runs My Paradise into Rock The House into Lemon8. It is 609.1 seconds long and contains the first two transitions whose duration and downbeat alignment have been approved by listening.

Not implemented

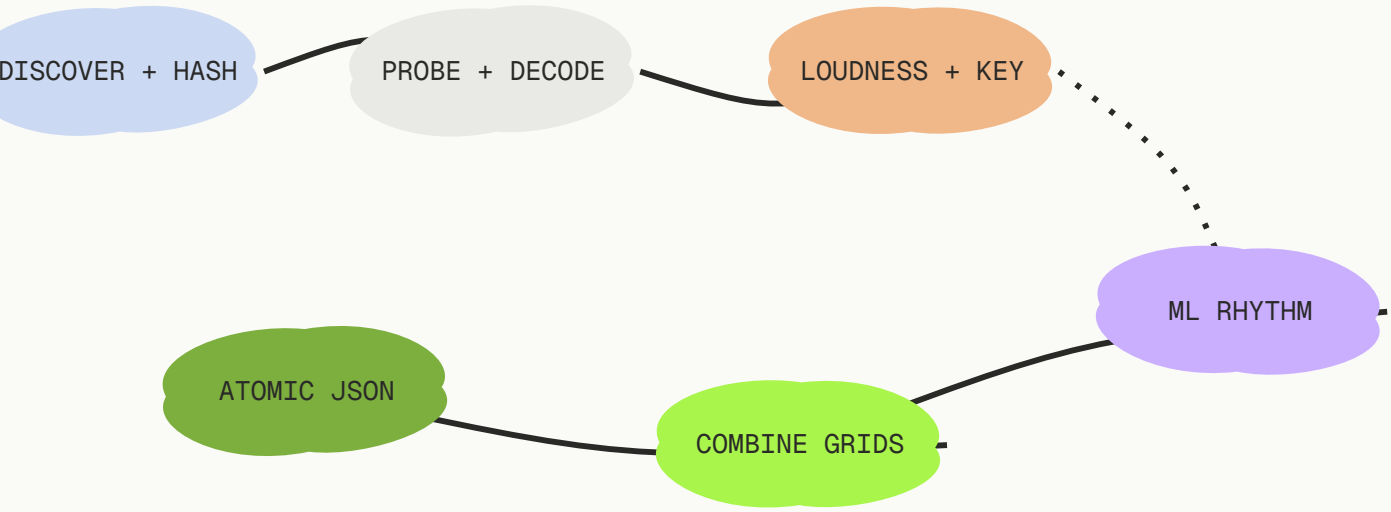
There is no natural-language planner, no automatic cue selection, no generic multi-point volume envelope, no delivery-mastered export, no stem-based mixing, and no broad full-corpus listening validation.

The honest label

This is a working manual-plan DJ-set engine with one successful reference render. It is substantially beyond a prototype script, but it is not yet a complete product that can take an arbitrary folder and brief without human planning.

Tempo, key, beat, downbeat, and section labels remain benchmark estimates. They are not promoted to facts merely because the pipeline completed.

HOW A TRACK BECOMES USABLE EVIDENCE



Every source is discovered, identified, and hashed before analysis. FFprobe and FFmpeg establish stream information and decoding health. Loudness and key are measured or estimated. All-In-One MLX supplies structural predictions, while Beat This runs on CPU to provide a second rhythm view. The results are reconciled into grids, downbeats, sections, and loop candidates, then written atomically with provenance, versions, warnings, and failures.

Why the pipeline is staged

Heavy work runs in a subprocess per track so the parent stays light. RSS, reclaimable memory, swap, and wired memory are watched. Start gates, recovery waits, caps, and selective reuse make retries survivable.

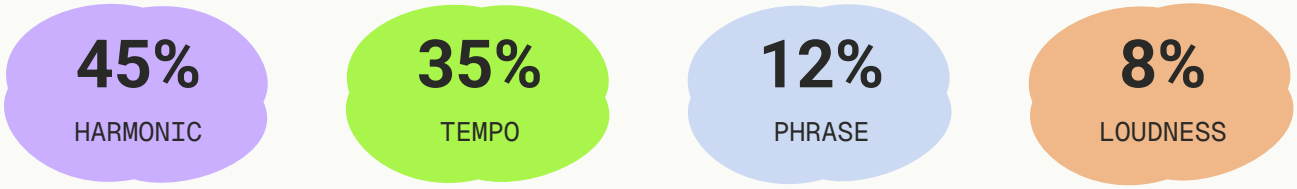
Why this matters

The 16 GB Mac encountered ENOSPC, SIGKILL, freezes, and memory kills before the corpus completed. Isolation, cleanup, checkpoints, telemetry, and reuse turned those failures into a repeatable pipeline.

AUTHORITY RULE

Facts and measurements stay distinct from model estimates. Human verification stays scoped to what was actually checked.

HOW THE ENGINE JUDGES A POSSIBLE NEXT TRACK



Compatibility is a planning heuristic, not a law of DJing. The combined score gives harmony the largest weight, followed by tempo, phrase opportunity, and loudness difference. If harmony falls outside policy, the combined score is capped at 0.15 so a convenient tempo match cannot disguise an unsupported harmonic relationship.

Harmonic policy

The default is Rekordbox Related 2: same Camelot key, relative major or minor, or one position around the wheel while staying on the same letter ring. Missing Camelot data is not treated as a match. Related 3 exists only as an opt-in relaxation.

Tempo policy

The engine prefers at most 2 percent incoming stretch, permits up to 4 percent, and rejects more. It never silently halves or doubles tempo. The outgoing track defines the shared BPM.

Phrase and loudness

Phrase scoring rewards shared usable 8- or 16-bar loops, then other shared loop sizes, then the mere presence of sections. Loudness scores are strongest within 3 LU and fall across wider gaps. These are practical gates, not aesthetic certainty.

Current planning limit

The tempo-wave order climbs and then descends, but the graph is fragmented. Dj Yellow at 120 BPM cannot reach Certified at 125 BPM within the 4 percent limit, and several harmonic links are weak. Choose a strong subset instead of forcing all ten tracks together.

THE MIXPLAN IS THE CONTRACT

THE PLAN DECLARES

The brief, allowed and excluded tracks, ordered IDs, segment start and end downbeats, gain, optional target BPM, transition adjacency, bar count, style, explanation, and planner identity.

THE VALIDATOR REJECTS

Unknown or duplicated IDs, unsafe paths, changed hashes, invalid cues, excessive gain or stretch, non-consecutive transitions, mismatched effective BPM, adjacent-grid drift above 70 ms, and implausible duration.

Once a plan passes, the renderer copies only the required source material, re-hashes it, extracts exact segments after input seeking, converts them to 44.1 kHz stereo 32-bit float PCM, applies bounded operations, assembles the set, and measures duration, LUFS, loudness range, and true peak. It then verifies source hashes again and publishes the WAV and RenderManifest atomically outside the music folder.

Why the validator matters

A future model can be imaginative about selection and storytelling without owning execution. Safety limits, source identity, cue bounds, timing consistency, and output placement remain deterministic and testable.

What reproducibility means

The same analysis artifact and MixPlan should lead to the same bounded processing decisions. The manifest records what was executed and what the result measured, rather than leaving the final WAV as an unexplained black box.

MANUAL TODAY, AUTOMATIC LATER

The current slice is intentionally manual. Proving the contract and renderer first prevents Step 3 from hiding audio or timing errors behind model behavior.

THE FIRST FAILURE FOUND THE RIGHT PROBLEM

V1 - REJECTED

Eight bars produced a 14.44-second overlap. Opposing whole-track volume fades and rounded-BPM timing made the transition rushed, weak, and about 104-114 ms early.

V2 - ACCEPTED

Sixteen exact detected bars produced 29.09 seconds. Drift measured 10 ms and 0 ms. house_eq_v1 gave the tracks clearer frequency roles, and the user accepted duration and alignment.

What house_eq_v1 actually does

During the first half of the overlap, incoming high frequencies rise while incoming bass stays out. Around the midpoint, low-frequency ownership moves from the outgoing track to the incoming track around a 180 Hz split, over no more than three seconds. Outgoing highs then leave during the second half. It is simple, but more deliberate than a triangular volume crossfade.

What the diagnosis proved

The downbeat-click renders allowed the stored grids to be heard directly in centered 30-second clips. The two rhythm analyzers already agreed strongly around the chosen transition regions, which shifted attention from detection failure to overlap construction. Exact detected-grid timing fixed the phase issue without pretending the downbeats were universally correct.

609.1 s	-13.81	3.1 LU	+1.04
TOTAL MIX	LUFS	LRA	DBTP

WHAT THE ENGINE CAN DO - AND ITS BOUNDS

INSPECT Track summaries, transition windows, and centered downbeat-click diagnostics.

SCORE AND ORDER Harmonic, tempo, phrase, loudness, combined score, and a greedy tempo wave.

CUE AND LOAD Exact extraction by seconds or downbeats; source hash is checked after copying.

LEVEL AND STRETCH Gain from -24 to +6 dB; pitch-preserving atempo with a 4 percent ceiling.

EQ AND FILTERING Bands at 120, 800, and 4000 Hz; low-cut from 80 to 300 Hz.

LOOP PCM repeated up to eight times with a 12 ms edge fade.

TRANSITION Linear or house_eq_v1; 4, 8, or 16 bars with exact-grid timing.

ASSEMBLE AND AUDIT Operation chains or a complete set, followed by WAV and manifest publication.

Not available yet

Generic volume envelopes, automatic cues, final-bus limiting, mastered formats, automatic MixPlan generation, language-model planning, and stem mixing. These are explicit gaps, not hidden capabilities.

FFmpeg 8.0.1 provides the current DSP. Rubber Band is not installed and remains a later quality comparison.

WHAT IS GOOD HERE - AND WHAT IS STILL THIN

Strong engineering

The separation between analysis, planning, validation, and execution is excellent for an MVP. Path containment, repeated hashing, output isolation, and atomic publication reduce the chance that experimentation damages music or creates ambiguous artifacts.

Provenance is also strong. Facts, measurements, estimates, user verification, proposals, and executed actions stay distinct. Recovery-aware analysis and 54 tests show a disciplined implementation rather than a loose collection of scripts.

Thin product evidence

Only one three-track sequence and two transitions have passed listening. Planning and cues remain manual. Transition vocabulary is narrow. Stretching has not been judged by ear. The output is unmastered and exceeds 0 dBTP.

Full analysis is expensive, the repository has no stable Git history, and no annotated truth set exists for judging changes to key, tempo, sections, or downbeats.

MY VERDICT

The deterministic core was proved before adding an LLM. The main weakness is no longer architecture; it is listening evidence.

WHAT COULD SLOW OR DISTORT THE NEXT PHASE

01

Final-bus safety

The accepted mix reaches +1.04 dBTP. Playback conversion or lossy encoding may clip, so delivery cannot be called finished until a deterministic ceiling and postcondition exist.

02

No version-control baseline

Every project file is currently untracked. The known-good state could be lost or confused with later experiments unless a deliberate first commit is created before major new work.

03

Sparse compatibility graph

Tempo and harmony constraints prevent a clean path through all ten tracks. The correct response is to choose a compatible subset, not weaken every rule merely to include everything.

04

Narrow listening proof

Two successful transitions do not validate every track, cue, arrangement, EQ shape, or stretched segment. A longer manual plan is the next generalization test.

05

No annotated ground truth

Model outputs can be compared with each other, but not yet against a durable human-reviewed set of cues, downbeats, keys, or sections. Analyzer optimization would therefore be difficult to measure.

06

Analysis cost

All-In-One can consume several GB of memory plus Metal-wired memory. Scaling to the roughly 120-track folder now would multiply cost and ambiguity before the product loop is finished.

Also unresolved: audible quality of 2-4 percent atempo stretching and the absence of an evaluation set for future planner behavior.

THE SHORTEST RESPONSIBLE ROUTE TO V1

1

Freeze the win

Create the first Git baseline while excluding source music, large renders, caches, and temporary work.

2

Make the output safe

Add deterministic final-bus headroom or limiting, record before-and-after measurements, and enforce a true-peak postcondition.

3

Test generalization

Build a five- or six-track manual set from the strongest compatible cluster. Preserve the accepted 16-bar timing and house_eq_v1 defaults.

4

Formalize listening

For every transition, record identity, timestamp, timing, phrasing, bass ownership, loudness balance, clashes, and pass or fail.

5

Fill only proven gaps

Add cue automation, richer envelopes, additional EQ moves, or stretch alternatives only when the listening record identifies a specific problem.

6

Start Step 3

After several manual plans work, let an LLM produce the same strict MixPlan and keep validation and rendering deterministic.

7

Scale later

Only then revisit analysis efficiency, the larger library, mixed genres, and interface work.

BEST NEXT BUILD

Final-bus safety is the smallest delivery gap. Preserve the accepted transition, measure it, then test a longer manual set.

WHERE THE SYSTEM LIVES

STATUS + INTENT

AGENTS.md holds the live takeover state and hard stops. PLAN.md carries the long product direction. V1_ENGINE_PLAN.md defines the implementation architecture. DECISION_LOG.md and EXPERIMENT_LOG.md preserve why decisions changed and what experiments actually showed.

CONTRACTS + CORPUS

docs/SCHEMAS.md defines serialized objects and authority labels. docs/CORPUS.md records the ten authorized source files, their SHA-256 hashes, and substitution policy. These contracts keep analysis, plans, manifests, and source identity explicit.

ANALYSIS CODE

src/dj_collection_set_builder/analyze.py orchestrates Step 1. track_worker.py, engines.py, and memory.py isolate heavy stages and protect resources. ingest.py, loudness.py, key_estimate.py, and rhythm.py produce the feature record.

PLANNING + RENDERING CODE

compatibility.py and story.py score and order relationships. summaries.py exposes compact planning context. render.py contains bounded audio primitives. set_renderer.py validates the MixPlan, assembles the set, measures it, and writes the manifest. downbeat_diagnostics.py creates click overlays.

ARTIFACTS + TESTS

examples/ contains the rejected v1 and accepted v2 plans. output/step1-collection.json is the completed analysis artifact. output/ also holds diagnostics, WAVs, and manifests. tests/ covers schemas, rhythm, memory, scoring, summaries, safety, rendering, and synthetic full-set execution.

Three entry points: dj-analyze builds the collection; dj-score inspects relationships and windows; dj-render creates diagnostics, clips, or a validated set. Environment: Apple Silicon, 16 GB RAM, Python 3.12.10, FFmpeg 8.0.1, Essentia 2.1b6.dev1389, All-In-One MLX, and Beat This CPU.

THE DURABLE MENTAL MODEL

1

Sources are immutable.

Every output lives elsewhere, and hashes prove that the originals survived.

2

Estimates are not facts.

Tempo, key, beats, downbeats, and sections remain fallible until verified at a precise scope.

3

Plans declare; code executes.

A future model may propose a MixPlan, but deterministic code owns validation and rendering.

4

The manual product works.

One three-track plan now has accepted transition duration and downbeat alignment.

5

Evidence comes before complexity.

Headroom and broader listening matter more now than extra intelligence or scale.

HARD STOPS

Do not modify the ten source MP3s, use the roughly 120-track folder, auto-render the corpus, start Step 3, weaken the memory floor, or swap analysis models merely to chase scale.

MAKE THE SOUND SAFE. MAKE THE TEST BIGGER. THEN ADD THE BRAIN.